UNIFORM SHEAFY TATE DOMAINS THAT ARE NOT STABLY UNIFORM

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Abstract. Work of Buzzard–Verberkmoes and, independently, Mihara shows that stable uniformity is a sufficient criterion for a Tate Huber ring to be sheafy, meaning that the structure presheaf of topological rings on its associated adic space is a sheaf. In this paper we construct examples of uniform sheafy rings which are not stably uniform, answering Question 7 of Kedlaya's *Nonarchimedean Scottish Book*. The examples were found in collaboration with ChatGPT 5.6 Sol. The first has been fully formalised in Lean 4; its proof rests on showing that a particular strict pullback of complete Tate rings is preserved by rational localisation. The second, whose bad localisation remains an integral domain, is proved here but has not yet been formalised.

1. Introduction

Buzzard and Verberkmoes proved that every stably uniform Tate Huber ring is sheafy and exhibited both uniform rings which are not stably uniform and uniform rings which are not sheafy [3]. Hansen and Kedlaya subsequently asked whether a uniform sheafy Huber ring must be stably uniform [4, Remark 3.16]; this is also Question 7 of Kedlaya's *Nonarchimedean Scottish Book* [11]. We give a negative answer in two forms.

We begin with the finite-jet pinching algebra. This is a nonnoetherian uniform integral domain for which a distinguished rational localisation is nonreduced, and hence is no longer uniform. Let k be a complete discretely valued nonarchimedean field, with valuation ring k°, uniformizer ϖ, and residue field $\widetilde{k}$. Put

$$L_0 = k^\circ\langle W, W^{-1}\rangle, \qquad B_0 = k^\circ\langle W, Q\rangle/(Q^2),$$

$$C_0 = L_0\langle Q\rangle, \qquad D_0 = L_0\langle Q\rangle/(Q^2),$$

where $k^\circ\langle W, W^{-1}\rangle$ means $k^\circ\langle W, V\rangle/(WV-1)$. There are natural maps $B_0 \to D_0$ and $C_0 \twoheadrightarrow D_0$. Define

$$A_0 = B_0 \times_{D_0} C_0, \qquad A = A_0[1/\varpi],$$

and endow A with the topology defined by the ring of definition A_0.

The main result is the following.

Theorem 1.1. *The ring A is a complete uniform Tate k-algebra and an integral domain, with $A^\circ = A_0$. The Huber pair (A, A°) is sheafy. However,*

$$A\left\langle \frac{W}{\varpi}\right\rangle \cong k\langle X, Q\rangle/(Q^2), \qquad X = \frac{W}{\varpi},$$

so A is not stably uniform.

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The proof uses a localisation theorem for Milnor squares of complete Tate rings. The terminology is classical rather than new. Milnor's patching theorem describes projective modules over a pullback ring with a surjective leg, and diagrams of this kind are now called Milnor squares [9, Section 2]. Milnor squares of complete Tate rings also appear explicitly in the analytic K-theory work of Kerz–Saito–Tamme [8, Section 3].

Definition 1.2. A commutative square of complete Tate k-algebras

$$\begin{array}{ccc} R & \longrightarrow & C \\ \downarrow & & \downarrow \\ B & \longrightarrow & D \end{array}$$

is a *strict Milnor square* if it is cartesian, the sequence of underlying topological k-vector spaces

$$0 \longrightarrow R \longrightarrow B \oplus C \xrightarrow{(b,c)\mapsto \bar{b}-\bar{c}} D \longrightarrow 0$$

is strict exact, and $C \to D$ is a strict surjection.

The adjective "strict" records the topology. In the present Banach setting it is not an additional hypothesis on the initial diagram: after rotating the square if necessary, every Milnor square of complete Tate k-algebras in the sense of Kerz–Saito–Tamme is strict by the nonarchimedean open mapping theorem [8, Lemma 3.1]. We retain the adjective because the substantive point proved here is that every completed rational localisation is again a Milnor square, with the corresponding strictness and bounded-denominator control, naturally under refinement. It follows that if B, C, D are sheafy, then so is R. Since Spa is contravariant, the associated square is a pushout in the full subcategory of affinoid adic spaces, but not in the category of all adic spaces; remark 6.3 explains the distinction and the failure for nonaffinoid targets. One may ask whether there is an example which does not rely on a rational localisation becoming nonreduced, and this is exactly what the second example achieves.

Both constructions emerged from an extended interaction with ChatGPT 5.6 Sol, after a number of unsuccessful attempts with earlier models. Since the first example was found in this way, we regarded machine-checked verification as an important part of the argument. The resulting Lean development, together with the precise correspondence between its definitions and those used here, is described in appendix A.

Remark 1.3 (Relation with earlier work)**.** It is difficult to determine precisely how an AI system arrived at such a construction, and hence to identify every human source that may have influenced it. We therefore record the closest related work which may have contributed ideas to the constructions below.

The closest precedents for the pathology are the examples of Buzzard–Verberkmoes and Mihara. Buzzard–Verberkmoes construct a uniform affinoid ring with a nonuniform rational localization and, separately, a uniform affinoid ring whose structure presheaf is not a sheaf [3, Propositions 17 and 18]. Mihara gives further examples of uniform Banach and affinoid rings for which rational localization loses uniformity, including a separate non-sheafy uniform affinoid ring [10]. Thus the phenomena "uniform but not stably uniform" and "uniform but not sheafy" were already known. What those examples did not supply was a proof that a non-stably-uniform example is nevertheless sheafy.

The pullback terminology also has a direct precedent. A cartesian square of rings with a surjective leg is a classical Milnor square, named for Milnor's theorem which patches projective modules from the two comparison rings [9, Section 2]. Kerz–Saito–Tamme already work explicitly with Milnor squares of complete Tate rings, construct compatible rings of definition by an open-mapping argument, and prove analytic K-theory excision for them [8, Section 3].

Thus neither the term "Milnor square", its complete-Tate version, nor the basic topological pullback mechanism is new here. We use the adjective "strict" to keep the topological exactness and bounded-denominator estimates visible; we have not found it used as standard terminology for this particular nonarchimedean situation. The new input needed for the present application is the graph–Koszul theorem showing that completed rational localisation preserves the square strictly and naturally, followed by the sheaf-transfer argument.

After applying the contravariant functor Spa, the ring pullback produces a pushout-shaped diagram of affinoid adic spaces; see remark 6.3. It is also analogous to a nonarchimedean pinching, but Temkin's pinching theorem concerns a finite gluing morphism, whereas in the present diagram $\mathrm{Spa}(D, D^\circ) \to \mathrm{Spa}(B, B^\circ)$ is a rational open immersion and is not finite [13]. Finally, the use of graph Koszul complexes to present rational localizations and control strict exactness over noetherian Tate algebras is closely related to the homological framework of Bambozzi–Kremnizer [1, Section 4]. These works provide the closest conceptual precedents for the ingredients of the construction; the finite-jet combination producing a uniform sheafy domain which is not stably uniform appears to be new.

The weighted-parity construction of section 7 is even closer in spirit to the infinite monomial-support examples of Buzzard–Verberkmoes and Mihara.

2. Conventions and notation

We follow Huber for Huber pairs, adic spectra, rational subsets, and rational localisation [5]; the bounded-denominator language for strictness follows Buzzard–Verberkmoes [3], and the graph presentation is also discussed by Hansen–Kedlaya [4].

Throughout, E denotes a complete Tate k-algebra. In the discretely valued setting of this paper, this means that E contains a topologically nilpotent unit and admits an open bounded k°-subalgebra E_0 such that

$$E = E_0[1/\varpi], \qquad \{\varpi^n E_0\}_{n\geq 0}$$

is a basis of neighbourhoods of zero. Such an E_0 is called a *ring of definition.* We always choose it closed, hence ϖ-adically complete. A subset of E is bounded if it is contained in $\varpi^{-r}E_0$ for some $r \geq 0$. If M is a complete topological k-vector space, an open bounded k°-submodule $M_0 \subset M$ will be called a lattice.

An element $x \in E$ is *power-bounded* if the set $\{x^n : n \geq 0\}$ is bounded. The ring of power-bounded elements is denoted E°; equivalently,

$$x \in E^\circ \iff x^n \in \varpi^{-r}E_0 \text{ for all } n \geq 0 \text{ and some } r \geq 0.$$

The ring E is *uniform* if E° is bounded, equivalently if

$$\varpi^r E^\circ \subset E_0$$

for some $r \geq 0$. It is *strongly noetherian* if $E\langle Z_1, \dots, Z_s\rangle$ is noetherian for every $s \geq 0$. These conditions do not depend on the chosen ring of definition.

A *ring of integral elements*, or *plus ring*, is an open subring $E^+ \subseteq E^\circ$ which is integrally closed in E; a *Huber pair* is a pair (E, E^+). We call (E, E^+) *sheafy* if its structure presheaf is a sheaf of complete topological rings. Thus on every finite rational cover the map to the matching-family equaliser must be a homeomorphism, not merely an algebraic isomorphism. Sections on arbitrary opens carry the projective-limit topology over rational subdomains. We call E *strongly sheafy* if every finite Tate extension $E\langle Z_1, \dots, Z_s\rangle$, equipped with its maximal plus ring, is sheafy.

We use the following bounded-denominator form of strictness. Let $u : M \to N$ be a continuous k-linear map and choose lattices $M_0 \subset M$ and $N_0 \subset N$. Then u is strict if and only if there exists $a \geq 0$ such that

$$\varpi^a \big(u(M) \cap N_0\big) \subset u(M_0). \tag{1}$$

For a surjection this becomes $\varpi^a N_0 \subset u(M_0)$. For an injection it says that the lattice induced from the target is bounded in the source. This is the criterion used by Buzzard–Verberkmoes [3, Lemma 2]. We use the term *strict Milnor square* in the sense of definition 1.2; the underlying algebraic terminology is standard [9, 8].

We now recall rational localization in the form that will be used below. We take every rational datum to have at least one numerator; an empty list may be padded by $f_1 = 0$. Let $\alpha = (f_1, \dots, f_m; g)$ be a rational datum in E, so the ideal generated by $g, f_1, \dots, f_m$ is open. Since E is a Tate k-algebra, this ideal is the unit ideal. After multiplying the entire datum by a common power of ϖ, we may assume that all entries lie in E_0.

The rational localization E_α is the separated ϖ-adic completion of $E[1/g]$ for the ring of definition generated by

$$E_0 \left[\frac{f_1}{g}, \dots, \frac{f_m}{g} \right].$$

It is independent of the ring of definition and of the presentation of the rational domain, and it is transitive under rational refinement. Equivalently, if

$$P_E = E\langle T_1, \dots, T_m \rangle = E_0\langle T_1, \dots, T_m \rangle[1/\varpi], \qquad r_i = gT_i - f_i,$$

then the ideal

$$J_{E,\alpha} := (r_1, \dots, r_m) = \operatorname{im}\left(P_E^m \xrightarrow{d_{1,E}} P_E \right) \subset P_E, \qquad d_{1,E}(u_1, \dots, u_m) = \sum_i u_i(gT_i - f_i), \tag{2}$$

is called the *graph ideal* associated with the rational datum α. When the datum is fixed, we abbreviate it to J_E. Before closedness has been proved, one must distinguish this algebraic ideal from its closure $\overline{J_{E,\alpha}}$; the rational localization is

$$E_\alpha \cong P_E / \overline{J_{E,\alpha}}. \tag{3}$$

The terminology comes from the equations themselves. After inverting g, the relations $gT_i - f_i = 0$ become

$$T_i = \frac{f_i}{g} \qquad (1 \leq i \leq m),$$

so they cut out the graph of the tuple $(f_1/g, \dots, f_m/g)$ in the relative affine m-space over $E[1/g]$. The Tate variables are power-bounded, so the same presentation records the rational inequalities $|f_i| \leq |g|$. These statements are standard: see Huber's universal property and presentation-independence results [5, (1.2), Proposition 1.3, Lemma 1.5]; the graph presentation is also recorded explicitly in [4, Definition 3.6].

We use the standard literature convention that a Tate ring is *stably uniform* when the completed localization associated with every genuine rational datum is uniform.

3. The global ring

In this section we identify the finite-jet algebra with a concrete support subring and prove that it is a complete uniform domain.

Set

$$L = L_0[1/\varpi], \quad B = B_0[1/\varpi], \quad C = C_0[1/\varpi], \quad D = D_0[1/\varpi].$$

The map $C_0 \to D_0$ has a continuous k°-linear section given by truncation in the Q-variable:

$$f_0 + Qf_1 \longmapsto f_0 + Qf_1.$$

Consequently

$$0 \longrightarrow A_0 \longrightarrow B_0 \oplus C_0 \longrightarrow D_0 \longrightarrow 0 \tag{4}$$

is exact, and A_0 is ϖ-adically complete. After inverting ϖ we obtain a strict Milnor square

$$\begin{array}{ccc} A & \longrightarrow & C \\ \downarrow & & \downarrow \\ B & \longrightarrow & D. \end{array}$$

In fact, the chosen rings of definition already make the integral row exact, so no powers of ϖ are lost in the defining square.

The map $B_0 \to D_0$ is injective. Projection to C_0 therefore identifies A_0 with the subring

$$A_0 = \left\{ f_0(W) + Qf_1(W) + Q^2 h(W, W^{-1}, Q) : f_0, f_1 \in k^\circ\langle W\rangle,\ h \in C_0 \right\}. \tag{5}$$

Equivalently, A_0 consists of the restricted series

$$\sum_{(a,b)\in S} c_{a,b} W^a Q^b, \qquad S = \{(a,b) \in \mathbf{Z} \times \mathbf{N} : b \le 1 \Rightarrow a \ge 0\}, \tag{6}$$

with $c_{a,b} \in k^\circ$. Here restricted means that, for every n, only finitely many coefficients are nonzero modulo ϖ^n. The set S is closed under addition, so this is a subring of C_0. The same description with coefficients in k gives $A \subset C$.

Write

$$\iota_B : A \to B, \qquad \iota_C : A \to C, \qquad \rho_B : B \to D, \qquad \rho_C : C \to D$$

for the structural maps. Although all topologies in this paper are specified by rings of definition, we will use once the coefficientwise Gauss norms whose unit balls are the displayed rings of definition. Give A the restricted Gauss norm from C. For $a = f_0 + Qf_1 + Q^2 h$, the coefficient formulas give

$$\|\iota_C(a)\|_C = \|a\|_A, \qquad \|\iota_B(a)\|_B \le \|a\|_A, \qquad \|\rho_B(b)\|_D \le \|b\|_B, \qquad \|\rho_C(c)\|_D \le \|c\|_C. \tag{7}$$

Thus every structural map is norm-nonincreasing. These estimates will be used only to verify preservation of power-bounded elements in theorem 6.2; they are not being used as an alternative definition of the topology.

Proposition 3.1. *The ring A is a complete uniform Tate k-algebra and an integral domain. Moreover,*

$$A^\circ = A_0.$$

Proof. For $0 \ne c \in C$, define

$$\nu(c) = \max\{n \in \mathbf{Z} : c \in \varpi^n C_0\}.$$

This maximum exists because C_0 is ϖ-adically separated and $C = C_0[1/\varpi]$. We claim that

$$\nu(cc') = \nu(c) + \nu(c') \qquad (c, c' \ne 0). \tag{8}$$

After multiplying c and c' by suitable powers of ϖ, it is enough to treat the case $c, c' \in C_0 \setminus \varpi C_0$. Their reductions are nonzero elements of

$$C_0/\varpi C_0 \cong \widetilde{k}[W, W^{-1}, Q].$$

This is a domain, so the product of the reductions is nonzero. This proves (8); in particular, C is a domain.

The support description gives
$$A_0 = A \cap C_0.$$
Every element of A_0 is power-bounded because A_0 is a subring. Conversely, let $a \in A \setminus A_0$. Then $\nu(a) < 0$, and by (8), $\nu(a^n) = n\nu(a)$. Hence the powers of a cannot all lie in any fixed $\varpi^{-r}A_0$. Thus a is not power-bounded. It follows that $A^\circ = A_0$, so A is uniform. Completeness follows from the completeness of A_0, and A is a domain because it is a subring of C. $\square$

4. A nonuniform rational localization

We next compute the rational chart which witnesses the failure of stable uniformity.

Consider the datum
$$\alpha_{\mathrm{in}} = (W; \varpi).$$
It is a genuine rational datum: the ideal (ϖ, W) is open (indeed ϖ is a unit in A). Let
$$A_{\mathrm{in}} = A\left\langle \frac{W}{\varpi} \right\rangle$$
be the corresponding rational localization on $|W| \le |\varpi|$, and write $X = W/\varpi$. By the ring-of-definition construction, A_{in} is the separated completion of the subring generated by A_0 and X.

Proposition 4.1. *There is a canonical isomorphism of complete Tate k-algebras*
$$A_{\mathrm{in}} \cong k\langle X, Q\rangle/(Q^2), \qquad W \longmapsto \varpi X.$$

Proof. Let
$$q\colon A_0[X] \longrightarrow A[X]/(\varpi X - W)$$
be the composite of the inclusion $A_0[X] \hookrightarrow A[X]$ with the quotient map, and set
$$G_0 := q(A_0[X]).$$
Since ϖ is a unit in A, evaluation at $X = W/\varpi$ gives a canonical isomorphism
$$A[X]/(\varpi X - W) \xrightarrow{\sim} A.$$
Under this identification,
$$G_0 = A_0[W/\varpi] \subset A;$$
it is the ring of definition whose separated ϖ-adic completion defines A_{in}. For $y \in Q^2C_0$ and every $n \ge 0$, the support condition gives $W^{-n}y \in A_0$. In G_0 one has
$$y = W^n(W^{-n}y) = \varpi^n X^n(W^{-n}y) \in \varpi^n G_0. \tag{9}$$
Thus Q^2C_0 maps to zero in the separated completion. After multiplying by a suitable power of ϖ, the same conclusion holds for every element of Q^2C.

Put
$$T_0 = k^\circ\langle X, Q\rangle/(Q^2), \qquad T = T_0[1/\varpi].$$
Using the unique expression in (5), define
$$f_0(W) + Qf_1(W) + Q^2h \longmapsto f_0(\varpi X) + Qf_1(\varpi X).$$
This sends A_0 into T_0, and it sends W/ϖ to X. Hence it extends to a continuous homomorphism $A_{\mathrm{in}} \to T$.

Conversely, X and Q belong to the ring of definition of A_{in}, and (9) gives $Q^2 = 0$ there. They therefore define a continuous homomorphism $T \to A_{\mathrm{in}}$. The two composites fix X and Q. On the dense image of G_0, the second composite differs from the identity only by an element of Q^2C_0, which is zero by (9). The two maps are therefore inverse. $\square$

Proposition 4.2. *The ring A is not stably uniform.*

Proof. In $T = k\langle X, Q\rangle/(Q^2)$, every element of the line kQ is nilpotent and hence power-bounded. This line is not bounded with respect to the ring of definition T_0: for every $r \geq 0$,

$$\varpi^r\left(\varpi^{-(r+1)}Q\right) = \varpi^{-1}Q \notin T_0.$$

Thus T is not uniform. By proposition 4.1, a genuine rational localization of the uniform ring A is nonuniform. □

5. Localization of the Milnor square

We now prove that every rational localization preserves the defining pullback. The argument is formulated entirely in terms of the chosen rings of definition.

5.1. **Graph Koszul complexes over affinoid algebras.** The following lemma will be used both at the noetherian comparison vertices and at the finite heads in section 7. Let E be an affinoid k-algebra, choose a noetherian ϖ-adically complete ring of definition $E_0 \subset E$, and let $f_1, \dots, f_m, g \in E_0$ generate the unit ideal in E. Put

$$P_E = E\langle T_1, \dots, T_m\rangle, \qquad P_{E,0} = E_0\langle T_1, \dots, T_m\rangle,$$

and set $r_i = gT_i - f_i$.

Lemma 5.1. *The Koszul complex $K_{P_E}(r_1, \dots, r_m)$ is exact in positive degrees. Every differential is continuous and strict onto its image, and every image is closed. In particular, if*

$$d_{1,E} : P_E^m \longrightarrow P_E, \qquad J_E = \operatorname{im}(d_{1,E}),$$

then J_E is closed. Moreover, there exists $h_E \geq 0$ such that

$$\varpi^{h_E}(J_E \cap P_{E,0}) \subset d_{1,E}(P_{E,0}^m). \tag{10}$$

There also exists $z_E \geq 0$ such that

$$\varpi^{z_E}\left(\ker(d_{1,E}) \cap P_{E,0}^m\right) \subset d_{2,E}\left(\textstyle\bigwedge^2 P_{E,0}^m\right). \tag{11}$$

Here positive-degree exactness means $\operatorname{Exact}(d_{q+1,E}, d_{q,E})$ *for every $q \geq 0$, equivalently exactness at the term of exterior degree $q+1$. It does not assert exactness at exterior degree zero and does not assert that $d_{1,E}$ is surjective onto P_E. For $m = 0$, the positive-degree assertion is vacuous.*

Proof. First work over $E[T_1, \dots, T_m]$. Let $\mathfrak{p}$ be a prime which does not contain all the r_i, and choose j with $r_j \notin \mathfrak{p}$. Then r_j is a unit after localization at $\mathfrak{p}$. If $e_1, \dots, e_m$ is the standard basis used to form the Koszul complex and d is the Koszul differential, define a degree-1 homotopy by

$$h(\omega) = r_j^{-1} e_j \wedge \omega.$$

The standard Koszul identity

$$d(e_j \wedge \omega) = r_j\omega - e_j \wedge d(\omega)$$

gives

$$dh + hd = \mathrm{id}\,.$$

Thus the identity chain map is chain-homotopic to the zero map; this is what it means here for the localized complex to be *contractible*. In particular it is exact: if $d\omega = 0$, then

$$\omega = (dh + hd)(\omega) = d(h\omega),$$

so every cycle is a boundary. Now let $\mathfrak{p}$ contain every r_i, and choose $1 = a_0 g + \sum_i a_i f_i$. In the local ring,

$$g\left(a_0 + \sum_i a_i T_i\right) = 1 + \sum_i a_i r_i.$$

The right-hand side is a unit, so g is a unit. After translating T_i by f_i/g, the sequence (r_i) differs by units from the coordinate sequence (T_i), which is regular over an arbitrary coefficient ring. Hence the polynomial Koszul complex is exact in positive degrees.

The ring E_0 is noetherian and ϖ-adically complete. More precisely, the canonical map

$$E[T_1, \dots, T_m] = E_0[T_1, \dots, T_m][1/\varpi] \longrightarrow E_0\langle T_1, \dots, T_m\rangle[1/\varpi] = P_E$$

is flat: $E_0[T_1, \dots, T_m]$ is noetherian, its ϖ-adic completion $E_0\langle T_1, \dots, T_m\rangle$ is flat over it by noetherian adic-completion flatness [12, Tag 00MB], and localization at ϖ preserves flatness. Flat base change therefore carries the positive-degree polynomial Koszul exactness to P_E.

Finally, P_E is a complete noetherian Tate ring. Huber's finite-module results imply that homomorphisms between finite P_E-modules are strict and that their images are closed [5, Lemmas 2.3 and 2.4]. Applying the bounded-denominator criterion (1) to $d_{1,E}$ and to $d_{2,E}$ onto $\ker(d_{1,E})$ gives (10) and (11). $\square$

5.2. **The graph ideals form a pullback.** Fix a rational datum

$$\alpha = (f_1, \dots, f_m; g)$$

in A. Multiplying all entries by a common power of ϖ, we assume $f_1, \dots, f_m, g \in A_0$. For $E \in \{A, B, C, D\}$ use the notation P_E, $P_{E,0}$, and $d_{1,E}$ introduced above, and abbreviate the graph ideal $J_{E,\alpha}$ to J_E.

The exact integral row (4) and the coefficientwise section of $C_0 \to D_0$ give an exact sequence

$$0 \longrightarrow P_{A,0} \longrightarrow P_{B,0} \oplus P_{C,0} \longrightarrow P_{D,0} \longrightarrow 0. \tag{12}$$

The map $P_{C,0} \to P_{D,0}$ is surjective, as are the induced maps on finite free modules and their exterior powers. Exactness is coefficientwise: a compatible restricted coefficient pair lifts to A_0, and the lifted coefficients still tend to zero ϖ-adically because (4) remains exact after multiplication by every power of ϖ.

Lemma 5.2. *The graph ideal J_A is closed in P_A, and the canonical map*

$$J_A \longrightarrow J_B \times_{J_D} J_C$$

is an isomorphism. Moreover,

$$0 \longrightarrow J_A \longrightarrow J_B \oplus J_C \longrightarrow J_D \longrightarrow 0 \tag{13}$$

is strict exact.

Proof. We first prove the algebraic pullback statement. Let $(x_B, x_C) \in J_B \times_{J_D} J_C$. Choose $u_B \in P_B^m$ and $u_C \in P_C^m$ with $d_{1,B}(u_B) = x_B$ and $d_{1,C}(u_C) = x_C$. Their difference in P_D^m lies in $\ker(d_{1,D})$. By lemma 5.1, it is $d_{2,D}(v_D)$ for some $v_D \in \bigwedge^2 P_D^m$. Lift v_D to $v_C \in \bigwedge^2 P_C^m$ and replace u_C by $u_C + d_{2,C}(v_C)$. The new pair of coefficient vectors has the same image in P_D^m, hence comes from $u_A \in P_A^m$ by (12). Then $d_{1,A}(u_A)$ maps to (x_B, x_C). This proves the asserted algebraic isomorphism.

We next verify bounded denominators. Choose exponents h_B, h_C, z_D as in lemma 5.1, and put $h = \max\{h_B, h_C\}$. Suppose that (x_B, x_C) belongs to the pullback and lies in $P_{B,0} \oplus P_{C,0}$. Choose $u_B \in P_{B,0}^m$ and $u_C \in P_{C,0}^m$ such that

$$d_{1,B}(u_B) = \varpi^h x_B, \qquad d_{1,C}(u_C) = \varpi^h x_C.$$

The difference $s = (u_B)_D - (u_C)_D$ belongs to $\ker(d_{1,D}) \cap P^m_{D,0}$. By (11), choose $v_D \in \bigwedge^2 P^m_{D,0}$ with $d_{2,D}(v_D) = \varpi^{z_D} s$. Lift v_D to $v_C \in \bigwedge^2 P^m_{C,0}$ and set

$$u'_C = \varpi^{z_D} u_C + d_{2,C}(v_C).$$

Then $(u'_C)_D = \varpi^{z_D}(u_B)_D$ and $d_{1,C}(u'_C) = \varpi^{h+z_D} x_C$. By (12), the compatible integral pair $(\varpi^{z_D} u_B, u'_C)$ comes from $u_A \in P^m_{A,0}$. Thus

$$\varpi^{h+z_D}\big((J_B \times_{J_D} J_C) \cap (P_{B,0} \oplus P_{C,0})\big) \subset d_{1,A}(P^m_{A,0}). \tag{14}$$

This proves strictness of the pullback identification. The fiber product $J_B \times_{J_D} J_C$ is closed in $P_B \oplus P_C$, because J_B and J_C are closed and the matching condition is closed. The integral row (12) identifies P_A strictly with the kernel of $P_B \oplus P_C \to P_D$. Together with (14), this shows that J_A is closed in P_A.

Finally, the map $J_B \oplus J_C \to J_D$ is surjective: lift a coefficient vector in P^m_D to P^m_C and apply d_1. If the element of J_D is integral, then (10) for $E = D$ shows that a fixed power of ϖ has an integral coefficient vector, which can be lifted to $P^m_{C,0}$. Thus this surjection is strict. Together with (14), this proves strict exactness of (13). □

Proposition 5.3. *For every rational datum α in A, the canonical sequence*

$$0 \longrightarrow A_\alpha \longrightarrow B_\alpha \oplus C_\alpha \longrightarrow D_\alpha \longrightarrow 0 \tag{15}$$

is strict exact, and $C_\alpha \to D_\alpha$ is a strict surjection. Equivalently,

$$A_\alpha \cong B_\alpha \times_{D_\alpha} C_\alpha.$$

These identifications are natural under rational refinement.

Proof. By lemma 5.2, all four graph ideals are closed. Hence the graph model (3) gives

$$E_\alpha = P_E/J_E \qquad (E = A, B, C, D).$$

The two exact rows (12) and (13) give algebraic exactness of (15).

Let $E_{\alpha,0}$ be the image of $P_{E,0}$ in E_α; this is a ring of definition. Since $P_{C,0} \to P_{D,0}$ is surjective, so is $C_{\alpha,0} \to D_{\alpha,0}$. Thus $C_\alpha \to D_\alpha$ is a strict surjection.

It remains to check strictness at the left. Let $x \in A_\alpha$ have images in $B_{\alpha,0}$ and $C_{\alpha,0}$. Choose representatives $b_0 \in P_{B,0}$ and $c_0 \in P_{C,0}$. Their difference

$$\delta = (b_0)_D - (c_0)_D$$

belongs to $J_D \cap P_{D,0}$. Choose h_D as in (10). There is $u_D \in P^m_{D,0}$ with $d_{1,D}(u_D) = \varpi^{h_D}\delta$. Lift u_D to $u_C \in P^m_{C,0}$. Then

$$\Big(\varpi^{h_D} b_0, \varpi^{h_D} c_0 + d_{1,C}(u_C)\Big)$$

is a compatible pair in $P_{B,0} \oplus P_{C,0}$, so it comes from $P_{A,0}$. Its class in A_α is $\varpi^{h_D} x$. Therefore

$$\varpi^{h_D}\Big(A_\alpha \cap (B_{\alpha,0} \oplus C_{\alpha,0})\Big) \subset A_{\alpha,0},$$

which is the bounded-denominator criterion for strictness.

All maps used above are induced by the original coefficient maps and the universal property of rational localization. Huber's presentation independence and transitivity therefore make the pullback identifications natural under rational refinement. □

6. Sheafiness

We isolate the elementary descent statement used to transfer sheafiness from the three comparison rings.

Lemma 6.1 (Sheaf transfer). *Let*

$$\begin{array}{ccc} R & \longrightarrow & C \\ \downarrow & & \downarrow \\ B & \longrightarrow & D \end{array}$$

be a commutative square of complete Tate Huber pairs. For $E \in \{B, C, D\}$, write

$$p_E : X_E = \mathrm{Spa}(E, E^+) \longrightarrow X_R = \mathrm{Spa}(R, R^+)$$

for the induced map. For a rational domain $U \subset X_R$, put $U_E = p_E^{-1}(U)$, and set $U_R = U$. Suppose that for every rational domain $U \subset X_R$ the section rings satisfy a natural strict exact sequence

$$0 \longrightarrow \mathcal{O}_R(U) \longrightarrow \mathcal{O}_B(U_B) \oplus \mathcal{O}_C(U_C) \longrightarrow \mathcal{O}_D(U_D) \longrightarrow 0. \tag{16}$$

If the Huber pairs associated with B, C, D are sheafy as complete topological rings, then the Huber pair associated with R is sheafy as a complete topological ring.

Proof. We first work on the rational basis. Let $U = \bigcup_{i=1}^n U_i$ be a finite rational covering. For $E \in \{R, B, C, D\}$, let

$$r_E : \mathcal{O}_E(U_E) \longrightarrow \prod_i \mathcal{O}_E((U_i)_E)$$

denote product restriction, using $U_R = U$ and $(U_i)_R = U_i$ as specified in the statement. The first map in (16) is a topological embedding, and the same is true after replacing U by any U_i. Since B and C are sheafy, r_B and r_C are topological embeddings. Naturality gives a commutative square

$$\begin{array}{ccc} \mathcal{O}_R(U) & \longrightarrow & \mathcal{O}_B(U_B) \oplus \mathcal{O}_C(U_C) \\ \downarrow r_R & & \downarrow r_B \oplus r_C \\ \prod_i \mathcal{O}_R(U_i) & \longrightarrow & \prod_i \big(\mathcal{O}_B((U_i)_B) \oplus \mathcal{O}_C((U_i)_C)\big). \end{array}$$

The top and right arrows are topological embeddings, as is the bottom arrow. It follows that r_R is a topological embedding.

Now let $x_i \in \mathcal{O}_R(U_i)$ be a matching family. Its images in the B- and C-section rings glue to elements

$$b \in \mathcal{O}_B(U_B), \qquad c \in \mathcal{O}_C(U_C).$$

Their images in $\mathcal{O}_D(U_D)$ agree after restriction to every $(U_i)_D$. Since the structure presheaf for D is separated, they agree globally. Exactness on U therefore gives $x \in \mathcal{O}_R(U)$ mapping to (b, c). Exactness on each U_i shows that $x|_{U_i} = x_i$, and uniqueness follows in the same way. Thus r_R identifies $\mathcal{O}_R(U)$ homeomorphically with the matching-family equalizer, which is closed in the finite product because all section rings are Hausdorff.

The underlying presheaf is therefore a sheaf by the sheaf-on-a-basis theorem [12, Tag 009O]. To retain the topology on arbitrary opens, let $V = \bigcup_\lambda V_\lambda$ be an open covering and let M be its matching-family equalizer with the subspace topology. For every rational $U \subset V$, quasi-compactness gives a finite rational cover of U subordinate to the V_λ. Restriction from M to that finite cover, followed by the rational-basis result, gives a continuous map $M \to \mathcal{O}_R(U)$, independent of the chosen subordinate cover. These maps are compatible under rational restriction. Since

$$\mathcal{O}_R(V) = \varprojlim_{U \subset V,\ U \text{ rational}} \mathcal{O}_R(U)$$

with its projective-limit topology, the inverse of the algebraic gluing map $\mathcal{O}_R(V) \to M$ is continuous. Hence the gluing map is a homeomorphism, as required. □

Theorem 6.2. *The Huber pair (A, A°) is sheafy.*

Proof. Give B, C, D their maximal rings of integral elements. The max/Gauss estimates (7) show that every structural map is norm-nonincreasing. Consequently it sends bounded subsets to bounded subsets; in particular, if x is power-bounded, then the set of powers of its image is the image of the bounded set $\{x^n : n \geq 0\}$. The structural maps therefore send power-bounded elements to power-bounded elements and define morphisms of Huber pairs.

For a rational domain $U \subset \mathrm{Spa}(A, A^\circ)$, its inverse image in each comparison spectrum is defined by the image of the same rational datum. By proposition 5.3, the corresponding section rings form the natural exact sequence required in lemma 6.1. Each of B, C, D is an affinoid k-algebra, hence is sheafy by Huber's theorem [5, Theorem 2.2]. The conclusion follows from lemma 6.1. □

Remark 6.3 (Affinoid versus global pushouts)**.** The coefficient descriptions of the maximal plus rings give

$$A^\circ = B^\circ \times_{D^\circ} C^\circ.$$

Indeed, a compatible pair which is power-bounded in B and C has its constant and linear Q-coefficients in both $k\langle W\rangle$ and L_0; their intersection is $k^\circ\langle W\rangle$, while the higher Q-tail is already integral in C_0. Together with the pullback topology, this says that the defining square is a pullback of complete Huber pairs.

Put $X_E = \mathrm{Spa}(E, E^\circ)$ for $E \in \{A, B, C, D\}$. Huber's affinoid mapping theorem identifies morphisms between affinoid adic spaces with continuous morphisms of the corresponding complete Huber pairs [5, Proposition 2.1]. Hence, for every affinoid adic space Z, the natural map

$$\mathrm{Hom}(X_A, Z) \longrightarrow \mathrm{Hom}(X_B, Z) \times_{\mathrm{Hom}(X_D, Z)} \mathrm{Hom}(X_C, Z)$$

is bijective. Thus the contravariant square

$$\begin{array}{ccc} X_D & \longrightarrow & X_C \\ \downarrow & & \downarrow \\ X_B & \longrightarrow & X_A \end{array}$$

is a pushout in the full subcategory of affinoid adic spaces.

This does not make it a pushout in the category of all adic spaces: there the same map must be bijective for an arbitrary, possibly nonaffinoid, target Z. In fact, uniqueness already fails for an explicit nonaffinoid target. Let

$$U_1 = \{x \in X_A : |Q^2(x)| \leq |\varpi(x)|\}, \qquad U_2 = \{x \in X_A : |1| \leq |W(x)|\}, \qquad U = U_1 \cup U_2.$$

The map $X_B \to X_A$ factors through U_1, because $Q^2 = 0$ in B, while $X_C \to X_A$ factors through U_2, because both W and W^{-1} are power-bounded in C and hence $|W| = 1$ on X_C.

The open subset U is proper. Use additive valuations with values in $\Gamma = \mathbf{Z}^3_{\mathrm{lex}}$ and, for a nonzero restricted series in the support description (6), put

$$v\left(\sum c_{a,b} W^a Q^b\right) = \min_{\mathrm{lex}}\{(v_k(c_{a,b}), b, a) : c_{a,b} \neq 0\}.$$

Restrictedness makes the minimum exist, and the initial-term argument for the lexicographic monomial order makes v multiplicative. The support condition makes it nonnegative on $A_0 = A^\circ$, while $v(\varpi) > 0$, so it is continuous and defines a point $x \in X_A$. Since

$$v(Q^2) = (0, 2, 0) < (1, 0, 0) = v(\varpi), \qquad v(W) = (0, 0, 1) > 0,$$

one has $|Q^2(x)| > |\varpi(x)|$ and $|W(x)| < 1$; hence $x \notin U$.

Adic spaces may be glued along open subspaces. Glue two copies of X_A along U and write $Z = X_A^{(1)} \cup_U X_A^{(2)}$. The two open immersions $j_1, j_2 : X_A \to Z$ agree on U, so they agree after precomposition with both $X_B \to X_A$ and $X_C \to X_A$, but they are distinct at the two unglued copies of x. The displayed Hom map is therefore not injective for this Z. Thus X_A is not the pushout in the full category of adic spaces. More generally, the same doubling argument shows that a cocone whose two legs factor through a proper open subspace of its vertex cannot be a pushout in that category. We do not know whether the original span admits some other, necessarily nonaffinoid, pushout object.

Proof of theorem 1.1. Uniformity and the domain property are proposition 3.1; sheafiness is theorem 6.2; and failure of stable uniformity is proposition 4.2. □

7. A rationally stably reduced example

The finite-jet construction above loses uniformity because separated completion creates a nilpotent. We now give a second construction in which the bad rational localization is an integral domain. The loss of uniformity is instead caused by an unbounded family of nonnilpotent power-bounded elements.

Definition 7.1. A complete Tate ring is *rationally stably reduced* if every finite iterated rational localization is reduced. This terminology is used only in the present paper; it is meant to avoid confusion with geometric reducedness after extension of the ground field.

Theorem 7.2. *There is a complete Tate k-algebra $\mathcal{A}$ with the following properties.*

1. *$\mathcal{A}$ is a uniform, nonnoetherian integral domain and $\mathcal{A}^\circ = \mathcal{A}_0$ for the ring of definition displayed below.*
2. *The Huber pair $(\mathcal{A}, \mathcal{A}^\circ)$ is strongly sheafy.*
3. *The ring $\mathcal{A}$ is rationally stably reduced.*
4. *The genuine rational localization*
$$\mathcal{B} = \mathcal{A}\left\langle \frac{W}{\varpi} \right\rangle$$
is an integral domain but is not uniform.

In particular, failure of stable uniformity need not be caused by a nilpotent in the bad rational localization.

7.1. **The weighted-parity algebra.** Let $I = \mathbf{N}_{>0}$. For a finite-support multi-index $\nu = (\nu_n)_{n\in I} \in \mathbf{N}^{(I)}$, put

$$\omega(\nu) = \sum_{\substack{n\geq 1 \\ \nu_n \text{ odd}}} n. \tag{17}$$

Consider the submonoid

$$S = \left\{(a, \nu) \in \mathbf{N} \times \mathbf{N}^{(I)} : a \geq \omega(\nu)\right\}. \tag{18}$$

The inequality

$$\omega(\nu + \mu) \leq \omega(\nu) + \omega(\mu) \tag{19}$$

shows that S is closed under addition. It is locally finite: a fixed element of S has only finitely many decompositions as a sum of two elements of S.

Let $k[S]$ be the monoid algebra with basis $W^a U^\nu$, where $U^\nu = \prod_n U_n^{\nu_n}$. Give it the Gauss norm

$$\left\| \sum_{(a,\nu)\in S} c_{a,\nu} W^a U^\nu \right\|_G = \sup_{a,\nu} |c_{a,\nu}|. \tag{20}$$

Its completion is

$$\mathcal{A} = \left\{ \sum_{(a,\nu)\in S} c_{a,\nu} W^a U^\nu : c_{a,\nu} \in k, \quad c_{a,\nu} \longrightarrow 0 \right\}, \tag{21}$$

where convergence to zero is over the discrete index set S. Local finiteness makes the coefficientwise convolution finite, and the submultiplicativity of the Gauss norm on $k[S]$ extends multiplication continuously to $\mathcal{A}$. Put

$$\mathcal{A}_0 = \left\{ \sum_{(a,\nu)\in S} c_{a,\nu} W^a U^\nu \in \mathcal{A} : c_{a,\nu} \in k^\circ \right\}. \tag{22}$$

Then $\mathcal{A}_0$ is the closed unit ball for (20), it is ϖ-adically complete, and $\mathcal{A} = \mathcal{A}_0[1/\varpi]$.

The finite stages admit a useful affinoid presentation. For $N \geq 1$, put

$$\mathcal{A}_N = \frac{k\langle W, Y_1, Z_1, \ldots, Y_N, Z_N\rangle}{(Y_n^2 - W^{2n} Z_n)_{1\leq n\leq N}}. \tag{23}$$

Lemma 7.3 (Isometric finite-stage normal form)**.** *There is an isometric decomposition*

$$\mathcal{A}_N \cong \bigoplus_{\epsilon\in\{0,1\}^N}^{\max} k\langle W, Z_1, \ldots, Z_N\rangle Y^\epsilon. \tag{24}$$

Under the substitution

$$Y_n \longmapsto W^n U_n, \qquad Z_n \longmapsto U_n^2, \tag{25}$$

this identifies $\mathcal{A}_N$ isometrically with the closed subalgebra of $k\langle W, U_1, \ldots, U_N\rangle$ supported on the monomials in (18) *involving only $U_1, \ldots, U_N$. The transition maps $\mathcal{A}_N \to \mathcal{A}_{N+1}$ are isometric, and*

$$\mathcal{A} = \widehat{\bigcup_{N\geq 1} \mathcal{A}_N}. \tag{26}$$

Proof. Successive division by the monic relations $Y_n^2 - W^{2n} Z_n$ reduces every class uniquely to a sum

$$\sum_{\epsilon\in\{0,1\}^N} f_\epsilon(W, Z_1, \ldots, Z_N) Y^\epsilon.$$

The division process does not increase the Gauss norm, so the quotient norm is at most the maximum of the norms of the f_ϵ. Under (25), the summand indexed by ϵ has U_n-exponent congruent to ϵ_n modulo 2. Different parity classes therefore have disjoint monomial supports. The image of the normal form consequently has Gauss norm exactly $\max_\epsilon \|f_\epsilon\|$. Since the substitution is norm-nonincreasing on the original Tate algebra, the norm of this image is at most the quotient norm. This proves both uniqueness and the isometry (24).

Writing $\nu_n = 2q_n + \epsilon_n$ gives the unique factorization

$$W^a U^\nu = W^{a-\omega(\nu)} \prod_{n=1}^{N} Y_n^{\epsilon_n} Z_n^{q_n} \tag{27}$$

whenever $a \geq \omega(\nu)$ and ν is supported in $\{1, \ldots, N\}$. Thus the finite-stage image is exactly the claimed support algebra. The same description proves that the transition maps are isometric. Finally, every finite set of allowed monomials is contained in one finite stage, while finite-support series are dense in the restricted series (21); this proves (26). □

Proposition 7.4. *The ring $\mathcal{A}$ is a complete uniform Tate k-algebra and an integral domain, with*

$$\mathcal{A}^\circ = \mathcal{A}_0.$$

Proof. The Gauss norm on the countable restricted Tate algebra $k\langle W, U_1, U_2, \ldots\rangle$ is multiplicative. Indeed, every nonzero restricted series attains its norm. After scaling two nonzero series to norm one, reduction modulo ϖ leaves two nonzero polynomials in $\widetilde{k}[W, U_1, U_2, \ldots]$, and their product is nonzero. Scaling back proves multiplicativity. The support algebra $\mathcal{A}$ embeds isometrically into this Tate algebra by (21), so its Gauss norm is also multiplicative and it is a domain.

If $x \in \mathcal{A}_0$, all powers of x remain in $\mathcal{A}_0$. If $x \notin \mathcal{A}_0$, multiplicativity gives $\|x^m\|_G = \|x\|_G^m$, which is unbounded. Hence $\mathcal{A}^\circ = \mathcal{A}_0$. Completeness and the Tate property follow from (21) and the topologically nilpotent unit ϖ. □

Proposition 7.5. *The ring $\mathcal{A}$ is not noetherian.*

Proof. For $m \geq 1$, let

$$I_m = (Z_1, \ldots, Z_m) \subset \mathcal{A}.$$

There is a compatible family of norm-nonincreasing maps on the finite stages, and hence a bounded homomorphism

$$\psi_m : \mathcal{A} \longrightarrow k\langle T\rangle,$$

which sends W and every Y_j to zero, sends Z_{m+1} to T, and sends every other Z_j to zero. The relations in (23) are preserved. The map ψ_m kills I_m but not Z_{m+1}. Therefore

$$I_1 \subsetneq I_2 \subsetneq I_3 \subsetneq \cdots,$$

so $\mathcal{A}$ is not noetherian. □

7.2. **A reduced nonuniform rational chart.** The datum $(W; \varpi)$ is a genuine rational datum: the ideal (W, ϖ) is open because ϖ is a unit of $\mathcal{A}$. Define the complete k°-algebra

$$\mathcal{B}_0 = \left\{ \sum_{\nu \in \mathbf{N}^{(I)},\ d \geq \omega(\nu)} b_{d,\nu} X^d U^\nu : b_{d,\nu} \in \varpi^{\omega(\nu)} k^\circ, \quad \varpi^{-\omega(\nu)} b_{d,\nu} \longrightarrow 0 \right\}, \tag{28}$$

with norm

$$\left\| \sum b_{d,\nu} X^d U^\nu \right\|_{\mathrm{in}} = \sup_{d,\nu} |\varpi|^{-\omega(\nu)} |b_{d,\nu}|. \tag{29}$$

The subadditivity (19) shows that $\mathcal{B}_0$ is a ring. Put $\mathcal{B} = \mathcal{B}_0[1/\varpi]$.

Proposition 7.6. *There is a canonical isometric isomorphism*

$$\mathcal{A}\left\langle \frac{W}{\varpi} \right\rangle \xrightarrow{\sim} \mathcal{B}, \qquad W \longmapsto \varpi X. \tag{30}$$

The graph ideal $(W - \varpi X)$ in $\mathcal{A}\langle X\rangle$ is closed, so no additional separated quotient is required.

Proof. On rings of definition define

$$q_0 : \mathcal{A}_0\langle X\rangle \longrightarrow \mathcal{B}_0, \qquad q_0(W^a X^c U^\nu) = \varpi^a X^{a+c} U^\nu.$$

For a target monomial define a k°-linear section by

$$s\big(b_{d,\nu} X^d U^\nu\big) = \varpi^{-\omega(\nu)} b_{d,\nu} W^{\omega(\nu)} X^{d-\omega(\nu)} U^\nu. \tag{31}$$

The divisibility and support conditions in (28) make this well-defined, and (29) shows that s is isometric. Moreover, $q_0 s = \mathrm{id}$.

It remains to identify the kernel. Let $M = W^a X^c U^\nu$ and put $r = a - \omega(\nu) \geq 0$. If $r > 0$, set

$$D(M) = W^{\omega(\nu)} X^c U^\nu \sum_{j=0}^{r-1} W^{r-1-j} (\varpi X)^j,$$

and set $D(M) = 0$ when $r = 0$. Then

$$M - s q_0(M) = (W - \varpi X) D(M). \tag{32}$$

Each basis monomial occurring in $D(M)$ is still allowed in $\mathcal{A}_0\langle X\rangle$, and all its scalar coefficients have norm at most one. Thus D is norm-nonincreasing on finite-support series and extends continuously to the c_0-completion. If $x \in \ker(q_0)$, then

$$x = x - s q_0(x) = (W - \varpi X) D(x).$$

The reverse inclusion is immediate, so

$$\ker(q_0) = (W - \varpi X)\mathcal{A}_0\langle X\rangle. \tag{33}$$

Since q_0 is a split surjection onto the complete ring $\mathcal{B}_0$, this kernel is closed. The quotient norm is exactly (29): boundedness of q_0 gives one inequality, and the isometric section gives the reverse inequality. Inverting ϖ proves (30) and the closedness assertion. □

Proposition 7.7. *The ring $\mathcal{B}$ is an integral domain but is not uniform.*

Proof. The weighted convergence condition in (28) implies ordinary restricted convergence, because $|b_{d,\nu}| \leq |\varpi|^{\omega(\nu)} |\varpi|^{-\omega(\nu)} |b_{d,\nu}|$. Hence coefficientwise inclusion gives an injective homomorphism

$$\mathcal{B} \hookrightarrow k\langle X, U_1, U_2, \ldots\rangle.$$

The target is a domain by the Gauss-norm argument used in proposition 7.4; therefore $\mathcal{B}$ is a domain.

Inside $\mathcal{A}$ write $Y_n = W^n U_n$, as in (25), and put

$$T_n = \varpi^{-n} Y_n = X^n U_n \in \mathcal{B}.$$

For $r \geq 0$, formula (29) gives

$$\|T_n^{2r}\|_{\mathrm{in}} = 1, \qquad \|T_n^{2r+1}\|_{\mathrm{in}} = |\varpi|^{-n}. \tag{34}$$

Thus every T_n is power-bounded. The family $(T_n)_{n\geq 1}$ is not bounded: if $\varpi^N T_n \in \mathcal{B}_0$, then the coefficient of $X^n U_n$ must be divisible by ϖ^n, which forces $N \geq n$. No fixed N works for all n. Hence $\mathcal{B}^\circ$ is unbounded and $\mathcal{B}$ is not uniform. □

7.3. **Small perturbations of rational data.** The finite-head argument rests on the following elementary stability result.

Lemma 7.8 (Small perturbation lemma)**.** *Let (E, E^+) be a complete Tate Huber pair and choose a closed ring of definition $E_0 \subseteq E^+$. Let $\alpha = (f_1, \ldots, f_m; g)$ be a rational datum whose entries lie in E_0. Write $d_0 = g$ and $d_i = f_i$ for $i \geq 1$. Suppose that for some $\ell \geq 0$ there are $a_0, \ldots, a_m \in E_0$ with*

$$\varpi^\ell = \sum_{j=0}^{m} a_j d_j. \tag{35}$$

Let $d'_j \in E_0$ satisfy

$$d'_j - d_j \in \varpi^{\ell+1} E_0 \qquad (0 \leq j \leq m), \tag{36}$$

and write $\alpha' = (f'_1, \ldots, f'_m; g')$. Then α' is a rational datum, α and α' define the same rational subset of $\mathrm{Spa}(E, E^+)$*, and their completed rational localizations are canonically isomorphic.*

Proof. Write $d'_j - d_j = \varpi^{\ell+1} h_j$ with $h_j \in E_0$. Then

$$\sum_j a_j d'_j = \varpi^\ell (1 + \varpi H), \qquad H = \sum_j a_j h_j \in E_0.$$

The element $1 + \varpi H$ is a unit of E_0, with inverse given by its convergent geometric series. Multiplying the a_j by this inverse gives an integral Bezout relation for the primed datum; in particular, α' is open.

Let x lie in the rational subset defined by α. Since $a_j \in E_0 \subseteq E^+$, relation (35) gives

$$|\varpi|^\ell \leq |g(x)|.$$

The perturbations have value at most $|\varpi|^{\ell+1} < |g(x)|$, so $|g'(x)| = |g(x)|$ and $|f'_i(x)| \leq |g'(x)|$. This proves one inclusion of rational subsets. Applying the same argument to the primed Bezout relation proves the reverse inclusion.

Let E_α be the first rational localization and put

$$q = \frac{\varpi^\ell}{g} = a_0 + \sum_i a_i \frac{f_i}{g} \in E_\alpha^\circ.$$

Then

$$\frac{g'}{g} = 1 + \varpi h_0 q.$$

The second summand is topologically nilpotent. After adjoining the finitely many power-bounded elements involved to a ring of definition, the geometric series for its inverse and all powers of that inverse lie in the same bounded subring. Thus g'/g is a 1-unit with power-bounded inverse, and

$$\frac{f'_i}{g'} = \frac{f_i/g + \varpi h_i q}{1 + \varpi h_0 q}$$

is power-bounded. The universal property gives a homomorphism $E_{\alpha'} \to E_\alpha$. Repeating the argument with the primed Bezout relation gives a homomorphism in the opposite direction. Both composites restrict to the identity on E; universal-property uniqueness makes them inverse. The isomorphism is therefore canonical. ∎

7.4. **Finite-head rational localization.** Fix $N \geq 1$. For a tail multi-index $\mu = (\mu_n)_{n>N} \in \mathbf{N}^{(\{n>N\})}$, write $\mu_n = 2q_n + \epsilon_n$ with $\epsilon_n \in \{0,1\}$ and put

$$e_\mu = \prod_{n>N} Y_n^{\epsilon_n} Z_n^{q_n}. \tag{37}$$

The finite-stage normal forms give an isometric Banach-module decomposition

$$\mathcal{A} \cong \widehat{\bigoplus_\mu}^{c_0} \mathcal{A}_N e_\mu. \tag{38}$$

Multiplication is determined by

$$e_\mu e_\lambda = W^{\omega(\mu)+\omega(\lambda)-\omega(\mu+\lambda)} e_{\mu+\lambda}. \tag{39}$$

Here ω is applied only to the tail coordinates. Projection to the coefficient of e_0 is a norm-nonincreasing algebra retraction

$$\rho_N : \mathcal{A} \longrightarrow \mathcal{A}_N. \tag{40}$$

Proposition 7.9 (Coefficientwise localization)**.** *Let α be a rational datum in $\mathcal{A}_N$, and put $P = (\mathcal{A}_N)_\alpha$. There is a canonical topological algebra isomorphism*

$$\mathcal{A}_\alpha \xrightarrow{\sim} \widehat{\bigoplus_\mu}^{c_0} P e_\mu. \tag{41}$$

It is natural for rational refinements represented in a finite head: if $P \to Q$ is the corresponding head restriction map, the induced restriction sends

$$\sum_\mu p_\mu e_\mu \longmapsto \sum_\mu (p_\mu|_Q) e_\mu. \tag{42}$$

The same statement holds after increasing N.

Proof. Let

$$R = \mathcal{A}_N\langle T_1, \ldots, T_m\rangle, \qquad I = (gT_i - f_i)_i \subset R.$$

Since $\mathcal{A}_N$ is affinoid, the graph ideal I is closed and the graph map $d_1 : R^m \to I$ is strict; this is the degree-zero conclusion of lemma 5.1. In particular, there is a constant C such that every $y \in I$ has a lift $x \in R^m$ with $\|x\| \leq C\|y\|$.

By (38),

$$\mathcal{A}\langle T_1, \ldots, T_m\rangle \cong \widehat{\bigoplus_\mu}^{c_0} R e_\mu.$$

Because the graph relations lie in the finite head, the graph differential acts coefficientwise. Its image is exactly

$$\widehat{\bigoplus_\mu}^{c_0} I e_\mu.$$

Indeed, the inclusion from left to right is immediate. Conversely, for a null family $(y_\mu)_\mu$ in I, choose coefficientwise lifts x_μ with $\|x_\mu\| \leq C\|y_\mu\|$; then $(x_\mu)_\mu$ is again a null family. Thus the graph ideal over $\mathcal{A}$ is closed and

$$\frac{\widehat{\bigoplus}_\mu^{c_0} R e_\mu}{\widehat{\bigoplus}_\mu^{c_0} I e_\mu} \cong \widehat{\bigoplus_\mu}^{c_0} (R/I) e_\mu \tag{43}$$

as complete topological modules and as algebras. Since $R/I = P$, this proves (41).

Every map used in (43) is induced by the canonical graph quotient. If a further rational datum lies in the head, its graph differential is again coefficientwise, so the same construction gives (42). Presentation independence and transitivity of rational localization extend this compatibility to iterated refinements and to a larger finite head. □

Corollary 7.10 (Finite-head presentation)**.** *Every rational localization of $\mathcal{A}$ is canonically isomorphic to a ring of the form*

$$\widehat{\bigoplus_{\mu}}^{c_0} Pe_\mu, \tag{44}$$

where P is a rational localization of some affinoid head $\mathcal{A}_N$. Under this identification the inclusions obtained by enlarging the head have norm-nonincreasing algebra retractions, and their union is dense.

Proof. Start with a rational datum in $\mathcal{A}$. Multiplying all entries by one power of ϖ does not change the rational subset or localization, so assume that the entries lie in $\mathcal{A}_0$. Choose an integral Bezout relation

$$\varpi^\ell = \sum_j a_j d_j, \qquad a_j \in \mathcal{A}_0.$$

By (26), for some N the entries d_j admit approximations $d'_j \in \mathcal{A}_N \cap \mathcal{A}_0$ satisfying $d'_j - d_j \in \varpi^{\ell+1}\mathcal{A}_0$. By lemma 7.8, the primed datum gives the same rational subset and the same completed localization. The primed integral Bezout relation produced in that lemma remains valid after applying the retraction ρ_N; hence the primed datum is a rational datum already in $\mathcal{A}_N$. Now apply proposition 7.9.

For $M \geq N$, put $P_M = (\mathcal{A}_M)_\alpha$. The same proposition, with head M, identifies the localization with the c_0-sum over the variables of index $> M$. Coefficient projection gives a norm-nonincreasing retraction to P_M, and finite tail truncation proves that $\bigcup_M P_M$ is dense. ☐

7.5. **Sheafiness.**

Theorem 7.11. *The Huber pair $(\mathcal{A}, \mathcal{A}^\circ)$ is strongly sheafy.*

Proof. We first prove the topological sheaf condition on the rational basis of $X = \mathrm{Spa}(\mathcal{A}, \mathcal{A}^\circ)$. Let $U \subset X$ be rational. By corollary 7.10, choose a head presentation

$$E = \mathcal{O}_X(U) \cong \widehat{\bigoplus_{\mu}}^{c_0} P_M e_\mu, \qquad P_M = (\mathcal{A}_M)_\alpha,$$

for some M. Let E_U^+ be the plus ring defining the rational subspace U; it need not equal the maximal ring E°. Choose the standard closed ring of definition $P_{M,0} \subset P_M^\circ$ furnished by the head graph presentation. The coefficientwise graph quotient gives a closed ring of definition

$$E_0 = \widehat{\bigoplus_{\mu}}^{c_0} P_{M,0} e_\mu \subseteq E_U^+.$$

The inclusion holds because E_U^+ is an open, hence closed, subring containing the integral graph algebra. After enlarging the head, finite tail truncation shows that the corresponding head lattices are dense in E_0.

Suppose that $U = \bigcup_{i=1}^r U_i$ is a finite rational cover. Choose rational data for the U_i inside the Huber pair (E, E_U^+). Scale their entries into E_0 and choose integral Bezout relations. After a common enlargement of the head, the density just noted and lemma 7.8 replace all these data by data lying in one head lattice $P_{M,0}$, without changing any U_i or its completed section ring. Applying the coefficient projection $E \to P_M$ to the integral Bezout relation supplied by lemma 7.8 shows that each new datum is rational already in P_M. Let

$$V_i \subset \mathrm{Spa}(P_M, P_M^\circ)$$

be the corresponding rational domains.

We verify that the V_i cover. The inclusion $P_M \to E$ and coefficient projection $E \to P_M$ are norm-nonincreasing and split each other. They therefore define morphisms of the maximal Huber pairs and a split surjection

$$\operatorname{Spa}(E, E^\circ) \longrightarrow \operatorname{Spa}(P_M, P_M^\circ). \tag{45}$$

The maximal-pair spectrum $\operatorname{Spa}(E, E^\circ)$ is a subspace of $\operatorname{Spa}(E, E_U^+) = U$. Hence the given cover restricts to a cover of $\operatorname{Spa}(E, E^\circ)$. On this subspace the restricted U_i are exactly the inverse images of the V_i. The surjectivity in (45) therefore implies that the V_i cover $\operatorname{Spa}(P_M, P_M^\circ)$. This avoids identifying U with the maximal-pair spectrum, which need not be correct at higher-rank points.

Write

$$P_{M,i} = \mathcal{O}(V_i), \qquad P_{M,ij} = \mathcal{O}(V_i \cap V_j).$$

Naturality in proposition 7.9 gives canonical identifications

$$\mathcal{O}_X(U_i) \cong \widehat{\bigoplus_\mu}^{c_0} P_{M,i} e_\mu, \qquad \mathcal{O}_X(U_i \cap U_j) \cong \widehat{\bigoplus_\mu}^{c_0} P_{M,ij} e_\mu, \tag{46}$$

under which every restriction map is coefficientwise.

The affinoid algebra P_M is sheafy, so its rational-cover equalizer is an isomorphism of Banach spaces

$$P_M \xrightarrow{\sim} \operatorname{Eq}\left(\prod_i P_{M,i} \rightrightarrows \prod_{i,j} P_{M,ij}\right). \tag{47}$$

Its inverse is bounded; fix a bound C. Let $(x_i)_i$ be a matching family in the section rings in (46), and write $x_i = \sum_\mu x_{i,\mu} e_\mu$. For each μ, equation (47) glues $(x_{i,\mu})_i$ to a unique $x_\mu \in P_M$, with

$$\|x_\mu\| \le C \max_i \|x_{i,\mu}\|. \tag{48}$$

Because the cover is finite and every coefficient family is null, the right side tends to zero. Thus $\sum_\mu x_\mu e_\mu$ is an actual element of the c_0-sum and gives the unique global section. The same bound proves that the restriction map is a topological embedding. Hence the structure presheaf satisfies the sheaf condition as complete topological rings on the rational basis.

For an arbitrary open cover, apply the projective-limit argument used in the last paragraph of the proof of lemma 6.1. Namely, every rational subdomain of the open set is quasi-compact and admits a finite rational cover subordinate to the given cover; the rational-basis result glues continuously on that subdomain, and uniqueness makes these sections compatible under rational restriction. The defining projective-limit topology on arbitrary-open sections then makes the inverse gluing map continuous. Thus the public all-open structure presheaf is a sheaf of complete topological rings.

Finally, for auxiliary Tate variables $V_1, \dots, V_s$ one has, isometrically,

$$\mathcal{A}\langle V_1, \dots, V_s\rangle \cong \widehat{\bigoplus_\mu}^{c_0} \mathcal{A}_N\langle V_1, \dots, V_s\rangle e_\mu. \tag{49}$$

The finite heads remain affinoid, the retractions and perturbation lemma are unchanged, and the preceding proof applies verbatim. Therefore every finite Tate extension is sheafy. $\square$

7.6. **Reducedness of all rational localizations.**

Lemma 7.12. *Let P be a reduced ring and let J be any index set. Define*

$$\mathcal{F}_J(P) = \prod_{\mu \in \mathbf{N}^{(J)}} P\, U^{\mu}, \tag{50}$$

with multiplication by coefficientwise finite convolution. Then $\mathcal{F}_J(P)$ is reduced.

Proof. For every finite subset $J_0 \subset J$, setting the variables outside J_0 equal to zero gives a homomorphism

$$\tau_{J_0} : \mathcal{F}_J(P) \longrightarrow P[[U_j : j \in J_0]].$$

These homomorphisms jointly detect all coefficients. A finite-variable formal power-series ring over a reduced ring is reduced: the injection $P \to \prod_{\mathfrak{p} \in \operatorname{Spec}(P)} P/\mathfrak{p}$ induces an injection of formal power-series rings into a product of domains. If an element of $\mathcal{F}_J(P)$ is nilpotent, every τ_{J_0} kills it; joint coefficient detection then kills the original element. □

Theorem 7.13. *Every finite iterated rational localization of $\mathcal{A}$ is reduced.*

Proof. By transitivity of rational localization, it is enough to treat one rational localization. Use corollary 7.10 to write it as

$$E \cong \widehat{\bigoplus_{\mu}}^{c_0} P e_{\mu}, \qquad P = (\mathcal{A}_N)_{\alpha}. \tag{51}$$

The affinoid algebra $\mathcal{A}_N$ is a domain by lemma 7.3. A rational localization of a reduced affinoid algebra is reduced [2, Corollary 7.3.2/10]; see also [6, Remark 1.2.16]. Thus P is reduced.

The element W is a non-zero-divisor in $\mathcal{A}_N$. Affinoid rational localization is algebraically flat [2, Corollary 7.3.2/6]; see also [7, Proposition 15.1.1, proof]. Tensoring $0 \to \mathcal{A}_N \xrightarrow{W} \mathcal{A}_N$ with P therefore shows that multiplication by W is injective on P. If $P = 0$, the conclusion is immediate, so assume otherwise.

Let $J = \{n > N\}$. Define

$$\Phi : E \longrightarrow \mathcal{F}_J(P), \qquad \Phi\left(\sum_{\mu} x_{\mu} e_{\mu}\right) = \sum_{\mu} W^{\omega(\mu)} x_{\mu} U^{\mu}. \tag{52}$$

The multiplication rule (39) makes Φ multiplicative. If $\Phi(\sum x_{\mu} e_{\mu}) = 0$, coefficient comparison gives $W^{\omega(\mu)} x_{\mu} = 0$ for every μ. The W-regularity of P then gives $x_{\mu} = 0$ for every μ, so Φ is injective. By lemma 7.12, the target is reduced; hence E is reduced. □

Proof of theorem 7.2. The uniform domain assertion is proposition 7.4, and nonnoetherianity is proposition 7.5. Strong sheafiness is theorem 7.11. Rational stable reducedness is theorem 7.13. Finally, propositions 7.6 and 7.7 identify the genuine rational chart $|W| \leq |\varpi|$ as a nonuniform integral domain. □

Remark 7.14 (Relation with the first construction)**.** The finite-jet and weighted-parity examples use different mechanisms. In the finite-jet ring, a whole tail becomes infinitely ϖ-divisible and the bad chart acquires a nilpotent. In the weighted-parity ring, every rational calculation can be moved into a noetherian finite head and then carried coefficientwise through a c_0-tail. Its bad chart remains a domain, and nonuniformity is witnessed by the unbounded family $(T_n)_{n \geq 1}$ of nonnilpotent power-bounded elements. The use of an infinite monomial support and of unbounded power-bounded directions is in the tradition of the examples of Buzzard–Verberkmoes and Mihara; the finite-head coefficientwise descent is the additional feature that makes sheafiness and rational stable reducedness accessible here.

Appendix A. Lean formalisation and verification

This appendix concerns the finite-jet construction of theorem 1.1; the weighted-parity construction of section 7 has not yet been formalised. Rather than reproduce the proofs, we record the three definitions which are most useful for comparing the formal statement with the mathematics in the paper. The complete development is pinned to commit `b007a4f3d` of AINTLIB.

Remark A.1 (Draft reproducibility note)**.** The commit used for this draft exists in the local AINTLIB repository, but is not currently reachable from the public GitHub repository. Thus the source hyperlinks in this appendix are publication targets, rather than links which can yet be followed. The commit should be restored under a permanent branch or tag before the paper is released.

The audit was run with Lean `4.33.0-rc1` and mathlib commit `fd1d54bc`. It built all 24 targeted modules and the 3307-job umbrella target, found no `sorry` or `admit` in the finite-jet development, and checked the axioms of the headline declarations. The full audit report records the commands and their output.

A.1. **Sheafiness.** For an open subset $V \subseteq \mathrm{Spa}(A, A^+)$, the type `limitSections V` is the projective limit of the sections over the rational subdomains contained in V. The all-open sheaf condition is formalised as follows; the source is `SheafyPair.lean`.

```
structure IsLimitSheaf (A : Type u) [CommRing A] [TopologicalSpace A]
    [IsTopologicalRing A] [PlusSubring A] [IsHuberRing A]
    [HasLocLiftPowerBounded A] : Prop where
  injective : ∀ {V : Opens ↑(Spa A A⁺)} {ι : Type u}
    {U : ι → Opens ↑(Spa A A⁺)}
    (hle : ∀ i, U i ≤ V)
    (_ : (V : Set ↑(Spa A A⁺)) ⊆ ⋃ i, (U i : Set ↑(Spa A A⁺)))
    {x y : ↑(limitSections V)},
    (∀ i, limitRestrict (hle i) x = limitRestrict (hle i) y) → x = y
  glue : ∀ {V : Opens ↑(Spa A A⁺)} {ι : Type u}
    {U : ι → Opens ↑(Spa A A⁺)}
    (hle : ∀ i, U i ≤ V)
    (_ : (V : Set ↑(Spa A A⁺)) ⊆ ⋃ i, (U i : Set ↑(Spa A A⁺)))
    (s : ∀ i, ↑(limitSections (U i))),
    (∀ i j,
      limitRestrict (inf_le_left (a := U i) (b := U j)) (s i) =
      limitRestrict (inf_le_right (a := U i) (b := U j)) (s j)) →
    ∃ x : ↑(limitSections V), ∀ i, limitRestrict (hle i) x = s i
  isEmbedding : ∀ {V : Opens ↑(Spa A A⁺)} {ι : Type u}
    {U : ι → Opens ↑(Spa A A⁺)}
    (hle : ∀ i, U i ≤ V)
    (_ : (V : Set ↑(Spa A A⁺)) ⊆ ⋃ i, (U i : Set ↑(Spa A A⁺))),
    Topology.IsEmbedding (limitRestrictProd hle)
```

The first field is the usual separation axiom: two sections which agree on a cover are equal. The second is the gluing axiom, with compatibility expressed by restricting to $U_i \cap U_j$. These are precisely the two algebraic parts of the ordinary sheaf condition. The final field remembers that we are working with topological rings. It says that restriction to the product over the cover is a topological embedding; together with the first two fields this identifies the sections on V, with their projective-limit topology, with the matching-family equaliser as a topological ring.

For a complete Tate ring, the user-facing definition installs a chosen valid ring of integral elements and invokes this all-open condition. The ring-level definition then quantifies over both the choice of completion and the plus ring. These definitions are in `SheafyRing.lean`.

```
def IsSheafyFor (Aplus : RingOfIntegralElements A) : Prop :=
  letI := Aplus.toPlusSubring
  haveI : IsRingOfIntegralElements (A⁺ : Subring A) := Aplus.2
  haveI : HasLocLiftPowerBounded A := hasLocLiftPowerBounded_faithful
  IsLimitSheaf A

def IsSheafyComplete : Prop :=
  ∀ Aplus : RingOfIntegralElements A, IsSheafyFor A Aplus

def IsSheafyTateRing : Prop :=
  ∀ (P : PairOfDefinition A)
    (Bplus : RingOfIntegralElements (CompletionModel A P)),
    IsSheafyFor (CompletionModel A P) Bplus
```

Thus `IsSheafyFor A Aplus` is the formal counterpart of saying that the Huber pair (A, A^+) is sheafy. The two local instances in its definition contain the bookkeeping needed to construct the structure presheaf; they do not add a mathematical hypothesis. The universal quantifiers in `IsSheafyTateRing` make the ring-level notion independent of the chosen pair of definition and ring of integral elements.

A.2. **Strong noetherianity.** The formal definition first constructs the restricted power-series ring. A coefficient function tends to zero along the cofinite filter exactly when, for every neighbourhood of zero, all but finitely many coefficients belong to that neighbourhood. The definitions below are from `RestrictedPowerSeries.lean`.

```
def MvPowerSeries.IsRestricted {k : ℕ} {A : Type*}
    [CommRing A] [TopologicalSpace A]
    (f : MvPowerSeries (Fin k) A) : Prop :=
  Tendsto (fun s : Fin k →₀ ℕ => MvPowerSeries.coeff s f)
    cofinite (nhds 0)

class IsStronglyNoetherian (A : Type*) [CommRing A]
    [TopologicalSpace A] [NonarchimedeanRing A] : Prop where
  isNoetherianRing_restricted : ∀ k : ℕ,
    IsNoetherianRing (restrictedMvPowerSeriesSubring k A)
```

The subring `restrictedMvPowerSeriesSubring k A` therefore consists of the usual restricted series in k variables, and is the formal model of $A\langle T_1, \ldots, T_k\rangle$. Requiring this ring to be noetherian for every k is exactly the standard definition of strong noetherianity used in the paper. The use of multivariate power series rather than an iterated one-variable construction is only an implementation choice.

A.3. **The finite-jet example.** We finally record the definition of the example itself. In the following code, `PowerSeries.Restricted R 1` is the radius-one Tate algebra in one variable over R, `RestrictedLaurent K` is $K\langle W, W^{-1}\rangle$, and `DualNumber R` is $R[Q]/(Q^2)$. The source is `FJP/Over/JetRings.lean`.

```
abbrev L : Type u := RestrictedLaurent K

abbrev JetC : Type u := PowerSeries.Restricted (L K) (1 : ℝ)
```

```
abbrev JetB : Type u :=
  DualNumber (PowerSeries.Restricted K (1 : ℝ))

abbrev JetD : Type u := DualNumber (L K)

noncomputable def jetSupport : Subring (JetC K) where
  carrier := {f |
    qCoeff K 0 f ∈ nonnegSubring K ∧
    qCoeff K 1 f ∈ nonnegSubring K}
  zero_mem' := by
    constructor <;> rw [qCoeff_zero] <;> exact zero_mem _
  one_mem' := by
    constructor <;> rw [qCoeff_one]
    · rw [if_pos rfl]; exact one_mem _
    · rw [if_neg one_ne_zero]; exact zero_mem _
  add_mem' := fun {f g} hf hg => by
    constructor <;> rw [qCoeff_add]
    · exact add_mem hf.1 hg.1
    · exact add_mem hf.2 hg.2
  neg_mem' := fun {f} hf => by
    constructor <;> rw [qCoeff_neg]
    · exact neg_mem hf.1
    · exact neg_mem hf.2
  mul_mem' := fun {f g} hf hg => by
    constructor
    · rw [qCoeff_zero_mul]
      exact mul_mem hf.1 hg.1
    · rw [qCoeff_one_mul]
      exact add_mem (mul_mem hf.1 hg.2) (mul_mem hf.2 hg.1)

abbrev JetA : Type u := ↑(jetSupport K)
```

The predicate `nonnegSubring K` cuts out $K\langle W\rangle$ inside $K\langle W, W^{-1}\rangle$. Consequently the carrier condition says that the coefficients of Q^0 and Q^1 have no negative powers of W, while the coefficients of Q^n for $n \geq 2$ are unrestricted. In other words, `JetA K` is precisely

$$\left\{ f_0(W) + Qf_1(W) + Q^2 h(W, W^{-1}, Q) \;\middle|\; \substack{f_0, f_1 \in K\langle W\rangle \\ h \in L\langle Q\rangle} \right\},$$

which is the support description of A used in equation (5) after inverting ϖ. The closure fields in `jetSupport` are the direct formal check that this support condition defines a subring.

The file also constructs the four maps in the Milnor square and proves `milnorRow_exact`: every compatible pair in `JetB K` and `JetC K` comes from a unique element of `JetA K`. Thus the concrete support subring is proved to realise the pullback $B \times_D C$; Lean does not merely assume that identification. The public endpoints, collected in `SheafyEndpoints`, include the following.

```
theorem finiteJet_isDomain : IsDomain (JetA K) :=
  inferInstance

theorem finiteJet_not_noetherian :
    ¬ IsNoetherianRing (JetA K) :=
  not_isNoetherianRing_JetA K

theorem finiteJet_isUniform_of_dvr
    [IsDiscreteValuationRing 𝒪[K]] :
```

```
    TopologicalRing.IsUniform (JetA K) :=
  finiteJet_isUniform K (Uniformizer.ofDVR K)

theorem finiteJet_not_stablyUniform_of_dvr
    [IsDiscreteValuationRing 𝒪[K]] :
    ¬ TopologicalRing.IsStablyUniform (JetA K) :=
  finiteJet_not_stablyUniform K (Uniformizer.ofDVR K)

theorem finiteJet_isSheafyTateRing_of_dvr
    [IsDiscreteValuationRing 𝒪[K]] :
    IsSheafyTateRing (JetA K) :=
  finiteJet_isSheafyTateRing K (Uniformizer.ofDVR K)
```

Here K is any complete nontrivially normed ultrametric field for which $\mathcal{O}[K] = \{x : \|x\| \leq 1\}$ is a discrete valuation ring. No local compactness, finiteness of the residue field, characteristic, or perfection assumption is used. The final declaration applies the preceding definition of sheafiness to the finite-jet ring; the development also proves, for every valid plus ring, the corresponding categorical sheaf statement for the genuine all-open structure presheaf with its projective-limit topology.

The graph–Koszul argument of lemma 5.1, the rational-chart calculation, and the transfer of sheafiness through the concrete Milnor square are too large to reproduce usefully here. They are contained respectively in the restricted-exactness file and the strictness and closed-image file, `FJP/Over/Chart.lean`, and `FJP/Over/SheafTransfer.lean`.

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